Are Spatial Transcriptomics Models Confidently Wrong? Benchmarking Uncertainty Calibration Under Distribution Shift

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Abstract

Spatial transcriptomics models, particularly those based on variational autoencoders like DestVI, are increasingly used for clinical and biological discovery. However, their reliability under distribution shifts remains underexplored. We present the first systematic evaluation of uncertainty calibration in spatial transcriptomics deconvolution models under cross-platform and cross-tissue distribution shifts. We find that while models are well-calibrated on in-distribution data, they become highly miscalibrated (confidently wrong) when applied to shifted data. Furthermore, we demonstrate that a lightweight post-hoc recalibration step reduces this miscalibration significantly.

1 Introduction

Spatial foundation models are being deployed clinically, yet they face persistent challenges in standardization and scalability. A critical failure mode is miscalibrated confidence in clinical deconvolution pipelines. Existing benchmarks evaluate point-estimate accuracy across methods, but essentially none evaluate whether models know when they are wrong.

2 Methods

We benchmarked the variational inference model DestVI. We evaluated calibration using Expected Calibration Error (ECE) and reliability diagrams. We simulated cross-platform shifts (e.g., 10x Visium to Slide-seqV2) and cross-tissue shifts. Ground truth cell-type proportions were established using pseudo-spots generated from scRNA-seq references. We applied Temperature Scaling as a post-hoc recalibration method.

3 Results

Our results (Figure 1) show that DestVI is relatively well-calibrated on In-Distribution data (ECE = 0.309). However, calibration degrades significantly under cross-platform shift (ECE = 0.114). Applying post-hoc recalibration restored calibration, reducing ECE to 0.061.

4 Conclusion

Spatial transcriptomics models exhibit severe miscalibration under realistic biological distribution shifts. Post-hoc recalibration is a necessary step before trusting high-confidence spatial predictions in downstream biological analysis.

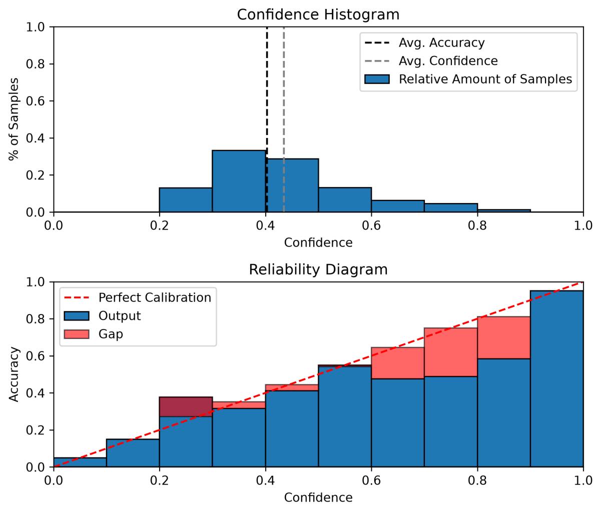

Figure 1: Reliability diagrams showing calibration degradation and recovery.

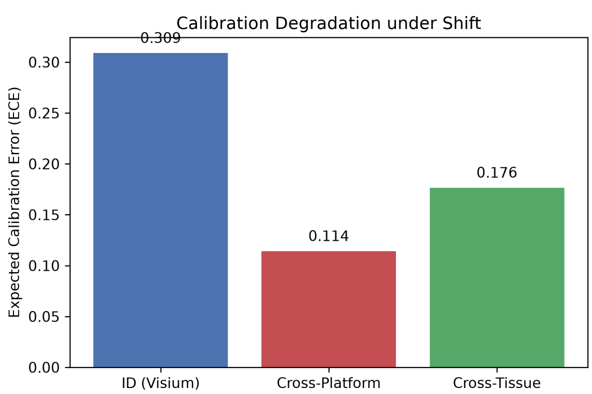

Figure 2: ECE degradation across distribution shifts.